

WordQuest — AI Agent Reference

A complete, current-state reference for every AI / logic agent powering WordQuest.

Each section explains **what the agent does**, **where it is wired up**, and **what the player actually sees** as a result.

Quick Map

#	Agent	Type	Lives In	Triggered By
1	levelGeneratorAgent	LLM (gpt-4o-mini)	<code>agents/levelGeneratorAgent.js</code>	Quick Play, Daily Challenge, Practice, 1v1 Battle, Pakistan Quest
2	difficultyAgent	Pure logic	<code>agents/difficultyAgent.js</code>	Quick Play (legacy fallback only)
3	refereeAgent	Pure logic	<code>agents/refereeAgent.js</code>	Word selection validation (server-side)
4	rewardAgent	Pure logic	<code>agents/rewardAgent.js</code>	Round-complete score breakdown
5	tutorAgent	LLM	<code>agents/tutorAgent.js</code>	"Explain this word" + AI Tutor screen
6	commentatorAgent	LLM	<code>agents/commentatorAgent.js</code>	Endpoint exists, mobile now uses local templates
7	coachAgent	LLM	<code>agents/coachAgent.js</code>	Game Over coach tip
8	chaalbaazAgent	Hybrid (logic + LLM)	<code>agents/chaalbaazAgent.js</code>	Quick Play, Practice — difficulty escalation

#	Agent	Type	Lives In	Triggered By
9	quizAgent	LLM	<code>agents/quizAgent.js</code>	Quiz mode
10	wordDetailAgent	LLM	<code>agents/wordDetailAgent.js</code>	Word Found card (meaning + example + syn/ant)
11	wordOfDayAgent	LLM	<code>agents/wordOfDayAgent.js</code>	Home screen banner
12	translateAgent	LLM	<code>agents/translateAgent.js</code>	Word detail translation button
13	lessonAgent	LLM	<code>agents/lessonAgent.js</code>	Learn tab lesson screens
14	learningPathAgent	Stateful logic + Supabase	<code>agents/learningPathAgent.js</code>	Home "Recommended for You" + after every round
15	guardrailAgent	Pure logic	<code>agents/guardrailAgent.js</code>	Safety pass on AI output (kids mode strict)
16	kidWordAgent	LLM	<code>agents/kidWordAgent.js</code>	Kids mode levels
17	kidQuestionAgent	LLM	<code>agents/kidQuestionAgent.js</code>	Kids mode questions

1. levelGeneratorAgent

What it does

Generates the actual word-search puzzle: an $N \times N$ grid of letters with target words placed horizontally, vertically, or diagonally (forwards or backwards). It picks the category, the words themselves, an emoji for the category, and a short fun fact.

Inputs: difficulty (easy / medium / hard), grid size, word count, language (English / Urdu), optional category override (Pakistan Quest pack, Practice mode), optional tier hints for CEFR-aligned word style.

Outputs: `{ category, categoryEmoji, words[], grid[][], wordPositions[], funFact }`

Where it is used

- **Quick Play** — every round at `POST /api/generate-level`
- **Daily Challenge** — same endpoint with `dailySeed` parameter for stable daily puzzles
- **Practice Mode** — `POST /api/practice/round` (category forced from a curated pool)
- **1v1 Battle** — `POST /api/battle/queue` generates a single shared level for both players
- **Pakistan Quest** — `POST /api/pakistan-quest/level` uses curated Pakistani vocabulary

What the player sees

The actual grid, the highlighted target words list, the category banner, the fun-fact ribbon at the bottom of the result screen.

2. difficultyAgent

What it does

Pure logic (no LLM call) that decides difficulty parameters (grid size, word count, time limit) from rolling player stats. Returns one of easy / medium / hard plus the numeric settings.

Where it is used

- **Quick Play legacy fallback** in `server.js` — only fires when no tier is supplied. Modern Quick Play uses the player's **tier** as the source of truth, so this is the fallback path.
- **Level Mode** — uses the same agent indexed by `levelNumber` to pull a locked difficulty table.

What the player sees

Indirectly — the grid size and timer they get. They never see this agent's reasoning string.

3. refereeAgent

What it does

Pure logic. Given the player's selected letter sequence and the target word list, decides whether the selection is a valid word find. Handles forward + reverse + all 8 directions.

Where it is used

- `POST /api/validate-word` — called every time the player completes a swipe across the grid in Quick Play / Daily / Level mode.

What the player sees

The instant "found!" animation + score popup OR the shake-rejection on a wrong selection.

4. rewardAgent

What it does

Pure logic. After a round, takes the words found, time left, hints used, perfect-clear flag, and produces the final score breakdown: base points, bonuses, penalties.

Where it is used

- `POST /api/round-complete` — fires from `RoundCompleteScreen.js` after every Quick Play / Daily / Level round.

What the player sees

The score breakdown card on the round-complete screen: "Words found", "Completion %", "Penalty", "New total".

Note: Current scoring also runs client-side via `letters + 2` base formula and `+25` perfect-clear bonus. The reward agent handles the canonical server-side tally.

5. tutorAgent

What it does

LLM-powered AI tutor. Two purposes:

1. **Word explanation** — give the meaning, example sentence, and a short lesson on a single word.
2. **Free chat tutor** — multi-turn AI Tutor conversation (Pro Max only, 30 messages / day cap).

Where it is used

- `POST /api/explain-word` — wired in `GameScreen.js` when player taps "Explain" on a found word
- **AI Tutor Screen** (`TutorScreen.js`) — chat interface with bobbing robot, daily cap counter, TTS speaker button

What the player sees

A clean explanation card in-game; or the full AI Tutor screen where they can ask follow-up questions and hear the tutor speak the reply.

6. commentatorAgent

What it does

LLM-powered live commentary lines for game events ("Nice combo!", "Halfway there!", "Watch the clock!").

Where it is used

- Endpoint `POST /api/commentary` still exists, but the **mobile client no longer calls it** — `GameScreen.js` uses local commentary templates for instant feedback (no network round-trip, no stacking lag). The agent is effectively **dormant** in production.

What the player sees

Local commentary popups during play. The LLM agent itself is currently bypassed.

7. coachAgent

What it does

LLM that reviews a finished round and produces one short coaching tip ("You found 7 / 10 — focus on diagonal scanning next time").

Where it is used

- `POST /api/coach` — called from `GameOverScreen.js` after the player loses or runs out of time.

What the player sees

The "Coach says..." card on the Game Over screen.

8. chaalbaazAgent (hybrid — recently revived)

What it does

The in-game "adversary". Two modes:

- **tune (pure logic)** — When the player is dominating their current tier or difficulty (streak ≥ 5 OR average 80%+ words found with time to spare), Chaalbaaz **bumps the difficulty one step**: +1 word, -10 sec, +1 grid cell. Never downgrades.
- **chat (LLM)** — English banter persona ("witty, slightly cocky, never insults"). Powers the standalone ChaalbaazChatScreen.

Where it is used

- **Quick Play** (`server.js` — tier branch) — tier sets the base difficulty, Chaalbaaz can escalate on top
- **Practice Mode** (`practiceApi.js`) — same escalation on top of the difficulty selector

What the player sees

The red "😏 **ADVERSARY ACTIVATED**" banner above the puzzle, with a one-line reason ("Streak of 6 — Chaalbaaz cranked up the heat."). The actual grid is bigger / harder.

9. quizAgent

What it does

LLM that generates a 20-question multiple-choice vocabulary quiz tailored to the player's tier / level. Each question has a correct answer + 3 plausible distractors + a short explanation.

Where it is used

- `POST /api/generate-quiz` — fired by **QuizScreen.js** at quiz start.

What the player sees

The full Quiz mode: question card, four options, instant correct/wrong feedback, final score screen with percentage.

10. wordDetailAgent

What it does

LLM that returns the rich detail for a single word: meaning, example sentence, synonym, antonym, optional translation.

Where it is used

- Routed via `tierApi.js` — called whenever the **Word Found card** appears (Quick Play, Daily, Level, Practice).

What the player sees

The card that pops up when they find a word, containing meaning + example + synonym + antonym. The full TTS read-aloud (speak everything in sequence) reads from this payload.

11. wordOfDayAgent

What it does

LLM that picks **one word of the day** with meaning + example + audio prompt, calibrated to the player's tier (Bronze gets simple words, Master gets advanced).

Where it is used

- Routed via `tierApi.js` — called once on **HomeScreen** load.

What the player sees

The "Word of the Day" banner card on the Home screen, with a speaker button to hear it read aloud.

12. translateAgent

What it does

LLM translator. Converts an English word/phrase into Urdu (Nastaliq script) and Roman Urdu.

Where it is used

- Routed via `tierApi.js` — used by **Word Detail card** when the player taps the translation toggle.

What the player sees

Urdu / Roman Urdu translation appearing alongside the English meaning.

13. lessonAgent

What it does

LLM that generates a structured lesson for the **Learn** tab: introductory text, key vocabulary, a short story / dialogue, and 3 practice questions.

Where it is used

- Mounted in `learnApi.js` — powers Lesson screens of the Learn path.

What the player sees

The Lesson detail page in the Learn flow — title, intro paragraph, highlighted vocab, story, and practice questions at the bottom.

14. learningPathAgent (Personalized Learning Path)

What it does

A **stateful** agent that maintains a per-player memory blob in Supabase (`wq_player_memory`).

After every round it:

- Updates rolling **EMA** metrics (avg score, avg words found, completion rate, time pressure, hint usage, diagonal speed, quiz accuracy)
- Tracks per-category accuracy
- Detects **weaknesses** (slow diagonals, hint overuse, low completion, time pressure, low quiz accuracy, weak categories)
- Generates a **next-3 recommendation** prescription tied to those weaknesses

When the home screen loads it returns the latest recommendations. **Cold start is tier-aware:** a new Elite-tier player gets Master Tier Climb / 1v1 / AI Tutor Deep Dive — *not* "Warm-up Practice".

Where it is used

- `POST /api/learning/session` — RoundCompleteScreen, QuizScreen, PracticeScreen all call this after a finished round
- `GET /api/learning/recommendations/:userId?tier=elite` — HomeScreen fetches on focus

What the player sees

The 🧠 **RECOMMENDED FOR YOU** card on Home, with weakness chips (`⚠️ Poor quiz accuracy` , `⚠️ Slow at diagonal words`) and three numbered, tappable prescriptions, each

routing to the right mode (Tutor / Quiz / Practice / 1v1 / Quick Play).

15. guardrailAgent (Safety Layer)

What it does

Pure-logic 5-layer safety filter that screens any AI output before it reaches the player. Layers:

1. Blocked terms (profanity, violence, NSFW)
2. Kid-mode strict block list (additional terms)
3. Always-block topics (regardless of mode)
4. Difficulty cap per age group
5. Final allow / block split with a reason string

Where it is used

- `POST /api/guardrail/check` — direct probe endpoint
- Conceptually a wrapper that every AI agent's response can be passed through. In kid mode it runs strictly.

What the player sees

Nothing visible when it passes. If content is blocked, the player gets a generic safe replacement / category swap. The block reason is logged to the admin Monitoring Dashboard.

16. kidWordAgent

What it does

LLM specialized for very young learners. Picks age-appropriate easy words (3–5 letters) for a chosen category, with friendly category emojis.

Where it is used

- Mounted in `kidsApi.js` — drives Kids Mode levels.

What the player sees

Kid-mode puzzles where every word is age-safe and short.

17. kidQuestionAgent

What it does

LLM that generates simple multiple-choice questions for Kids mode (e.g. "Which one is a fruit? 🍎 🚗 🐶 ⚽").

Where it is used

- Mounted in `kidsApi.js` — drives Kids Mode quizzes.

What the player sees

Kid-mode quiz screen with large emojis and simple questions.

Monitoring & Observability

Every agent call is logged through `utils/supabaseLogger.js` to the `wq_agent_logs` table. Each row captures:

- `agent` (name) · `status` (ok / error) · `durationMs` · `model` used
- `prompt` + `response` snapshots
- `meta` : userId, tool / reason / fallback pills, agent-specific metadata

The **Admin Monitoring Dashboard** (`adminDashboard.js`) shows this live with three tabs:

- Overview** — recent agent calls with tool / reason / fallback pills
- Pipelines** — session-grouped traces (silent refresh, won't kick you off the page)
- User Drill-down** — per-user agent history

Summary by Mode

Game Mode	Agents involved
Quick Play	levelGenerator · difficulty (fallback) · referee · reward · wordDetail · coach · chaalbaaz · learningPath · guardrail
Daily Challenge	levelGenerator · referee · reward · wordDetail · learningPath · guardrail

Game Mode	Agents involved
Practice	levelGenerator · referee · chaalbaaz · learningPath · wordDetail · guardrail
1v1 Battle	levelGenerator · referee (client) · learningPath · guardrail
Quiz	quiz · learningPath · guardrail
AI Tutor	tutor · guardrail
Pakistan Quest	levelGenerator (curated pack) · wordDetail · translate · guardrail
Kids Mode	kidWord · kidQuestion · guardrail (strict)
Learn Tab	lesson · guardrail
Home Screen	wordOfDay · learningPath

Document generated from the live codebase. For any agent's exact prompt or implementation detail, see the source file in [wordquest/backend/agents/](#).